the overlying sediment packages led to the apparent evolution of the domal inversion through the Eocene[19,25]. The estimated 200–300 m of crestal erosion during this phase[25] agrees well with the predicted relative magnitudes of erosion for the primary and secondary inversion (Fig. 2). Relaxation of compressional induced stresses by ductile flow processes in the softer parts of the lithosphere may have contributed to the Palaeogene evolution[9–11]. However, the plate-wide synchroneity of the Middle Palaeocene inversions indicates a sudden plate-wide release of inversion potential, a significant fraction of which must therefore have been maintained in compressionally over-deepened elastic flexures.

An erosional phase in the southern North Sea around the Eocene–Oligocene boundary has been explained by a Pyrenean phase of compression[3,19,25–27], but may not be easy to distinguish from the erosional effects of the global sea level fall at that time[29].

It is widely accepted that flexural isostasy provides a causal relation between in-plane stress variations, differential vertical movements of the surface of the lithosphere and changes in the depositional patterns of the sedimentary sequences[13]. Application of this concept requires knowledge about the initial lithospheric deflection, which generally is not available away from topographic loads in the interior of continents. The central regions of rift basins, however, tend to become positively loaded during the rifting and thermal sag phase and therefore develop an intrinsic downward flexure, which in some cases has been revealed by accelerated subsidence[15] in the incipient stage of compressional inversion. This pre-loading of rift basins promotes the formation of the asymmetric, primary marginal troughs during compressional inversion, and contributes positively to the over-deepening of the flexure that drives the relaxation inversion.

The existence of a positive load of the lithosphere along inversion ridges allows for prediction of the flexural response of the lithosphere to in-plane stress variations. Application of this concept to key European inversion structures allows us to explain the Middle Palaeocene change of inversion style, the synchroneity of the event across the European plate, the relative magnitude of the Late Cretaceous and Palaeocene erosion phases (when estimates exist) and the control exerted by the inversion movements on the post-Danian Palaeocene deposits.

The surprising conclusion that relaxation of in-plane tectonic stress, rather than increased Alpine compression, explains the European Middle Palaeocene inversions calls for reconsideration of the sources of the stresses affecting the interior of the European plate. Candidates for producing a plate-scale change of tectonic setting at this time involving forces in the range of 10^{12} N m^{-1} include uplift of the North Atlantic lithosphere by the Iceland plume[6] and the drop in north–south convergence rate between Africa and Europe[7]. □

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1. Voigt, E. Über Randtröge vor Schollenrändern und ihre Bedeutung im Gebiet der Mitteleuropäischen Senke und angrenzender Gebiete. *Z. Deutsch. Geol. Gesell.* **114,** 378–418 (1962).
2. Ziegler, P. A. Late Cretaceous and Cenozoic intra-plate compressional deformations in the Alpine foreland – a geodynamic model. *Tectonophysics* **137,** 389–420 (1987).
3. Ziegler, P. A. *Geological Atlas of Western and Central Europe* 2nd edn (Shell International Petroleum Maatschappij B.V., Geological Society Pubishing House, Bath, 1990).
4. Ziegler, P. A., Cloetingh, S. & van Wees, J. D. Dynamics of intra-plate compressional deformation: the Alpine foreland and other examples. *Tectonophysics* **252,** 7–22 (1995).
5. Ziegler, P. A., van Wees, J. D. & Cloetingh, S. Mechanical controls on collision-related compressional intraplate deformation. *Tectonophysics* **300,** 103–129 (1998).
6. White, N. J. & Lovell, B. Measuring the pulse of a plume with the sedimentary record. *Nature* **387,** 888–891 (1997).
7. Rosenbaum, G., Lister, G. S. & Duboz, C. Relative motions of Africa, Iberia and Europe during Alpine orogeny. *Tectonophysics* **359,** 117–129 (2002).
8. Erlström, M., Thomas, S. A., Deeks, N. & Sivhed, U. Structure and tectonic evolution of the Tornquist Zone and adjacent sedimentary basins in Scania and the southern Baltic Sea area. *Tectonophysics* **271,** 191–225 (1997).
9. Nielsen, S. B. & Hansen, D. L. Physical explanation of the formation and evolution of inversion zones and marginal troughs. *Geology* **28,** 875–878 (2000).
10. Hansen, D. L., Nielsen, S. B. & Lykke-Andersen, H. The post-Triassic evolution of the Sorgenfrei-Tornquist Zone – results from thermo mechanical modelling. *Tectonophysics* **328,** 245–267 (2000).
11. Hansen, D. L. & Nielsen, S. B. Why rifts invert in compression. *Tectonophysics* **373,** 5–24 (2003).
12. Clemmensen, A. & Thomsen, E. Palaeoenvironmental changes across the Danian-Selandian boundary in the North Sea Basin. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **219,** 351–394 (2005).
13. Lambeck, K., Cloetingh, S. & McQueen, H. Intraplate stress and apparent changes in sea level: the basins of Northwestern Europe. *Can. Soc. Petrol. Geol. Mem.* **12,** 259–268 (1987).
14. Gregersen, S., Voss, P. & TOR Working Group. Summary of project TOR: delineation of a stepwise, sharp, deep lithosphere transition across Germany–Denmark–Sweden. *Tectonophysics* **360,** 61–73 (2002).
15. Dadlez, R., Narkiewicz, M., Stephenson, R. A., Visser, M. T. M. & Van Wees, D.-J. Tectonic evolution of the Mid-Polish Trough: modelling implications and significance for central European geology. *Tectonophysics* **252,** 179–195 (1995).
16. Lamarche, J., Scheck, M. & Lewerenz, B. Heterogeneous inversion of the Mid-Polish Trough related to crustal architecture, sedimentary patterns and structural inheritance. *Tectonophysics* **373,** 141–159 (2003).
17. Vejbæk, O. V. & Andersen, C. Post mid-Cretaceous inversion tectonics in the Danish Central Graben. *Bull. Geol. Soc. Denmark* **49,** 129–144 (2002).
18. Clausen, O. R. & Korstgård, J. A. Tertiary tectonic evolution along the Arne-Elin Trend in the Danish Central Trough. *Terra Nova* **5,** 233–243 (1993).
19. de Jager, J. Inverted basins in the Netherlands, similarities and differences. *Neth. J. Geosci.* **4,** 355–366 (2003).
20. Baldschuhn, R., Best, G. & Kockel, F. N. in *Generation, Accumulation and Production of Europe's Hydrocarbons* (ed. Spencer, A. M.) 149–159 (Spec. Publ. 1, European Association of Petroleum Geoscientists, Oxford Univ. Press, Oxford, 1991).
21. Kockel, F. Inversion structures in Central Europe—Expressions and reasons, an open discussion. *Neth. J. Geosci.* **82,** 367–382 (2003).
22. Lake, S. D. & Karner, G. G. The structure and evolution of the Wessex Basin, southern England: an example of inversion tectonics. *Tectonophysics* **137,** 347–378 (1987).
23. Jolley, D. W. Palynostratigraphy and depositional history of the Paleocene Ormesby/Thanet depositional sequence set in southeastern England and its correlation with continental West Europe and the Lista Formation, North Sea. *Rev. Palaeobot. Palynol.* **99,** 265–315 (1998).
24. Mortimore, R. N. & Pomerol, B. Upper Cretaceous tectonic phases and end Cretaceous inversion in the Chalk of the Anglo-Paris Basin. *Proc. Geol. Ass.* **108,** 231–255 (1997).
25. Worum, G. & Michon, L. Implications of continuous structural inversion in the West Netherlands Basin for understanding controls on Palaeogene deformation in NW Europe. *J. Geol. Soc. Lond.* **162,** 73–85 (2005).
26. de Lugt, I. R., van Wees, J. D. & Wong, Th. E. in *Dynamics of Sedimentary Basin Inversion: Observations and Modelling* (eds Nielsen, S. B. & Bayer, U.) *Tectonophysics* **373** (Spec. Iss.), 141–159 (2003).
27. van Hoorn, B. Structural evolution, timing and tectonic style of the Sole Pit inversion. *Tectonophysics* **137,** 239–284 (1987).
28. Steurbaut, E. High-resolution holostratigraphy of Middle Paleocene to Early Eocene strata in Belgium and adjacent areas. *Palaeontographica Abt. A* **247,** 91–156 (1998).
29. Zachos, J., Pagani, M., Sloan, L., Thomas, E. & Billups, K. Trends, rhythms, and aberrations in global climate 65 Ma to present. *Science* **292,** 686–693 (2001).
30. Vinken, R. *et al.* The Northwest European Tertiary Basin. Isopach map of Middle-Late Paleocene deposits. *Geol. Jb.* **100,** 1–508 (1988).
31. Dadlez, R., Marek, S. & Pokorski, J. *Geological Map of Poland Without Cainozoic Deposits* (Polish Geological Institute, Warsaw, 2000).

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Palaeomagnetism of the Vredefort meteorite crater and implications for craters on Mars

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Magnetic surveys of the martian surface have revealed significantly lower magnetic field intensities over the gigantic impact craters Hellas and Argyre than over surrounding regions[1]. The reduced fields are commonly attributed to pressure demagnetization caused by shock waves generated during meteorite impact[2,3], in the absence of a significant ambient magnetic

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field. Lower than average magnetic field intensities are also observed above the Vredefort meteorite crater in South Africa, yet here we show that the rocks in this crater possess much higher magnetic intensities than equivalent lithologies found elsewhere on Earth. We find that palaeomagnetic directions of these strongly magnetized rocks are randomly oriented, with vector directions changing over centimetre length scales. Moreover, the magnetite grains contributing to the magnetic remanence crystallized during impact, which directly relates the randomization and intensification to the impact event. The strong and randomly oriented magnetization vectors effectively cancel out when summed over the whole crater. Seen from high altitudes, as for martian craters, the magnetic field appears much lower than that of neighbouring terranes, implying that magnetic anomalies of meteorite craters cannot be used as evidence for the absence of the planet's internally generated magnetic field at the time of impact.

The two-billion-year-old[4] Vredefort structure (Fig. 1) represents the vestiges of an immense (>300 km original diameter) impact crater, where rebound and subsequent erosion have unearthed 1,000 km^2 of well-exposed Archaean crystalline basement. As the Earth's largest and oldest known impact structure, Vredefort serves as the best terrestrial analogue for the large martian craters. Shock deformation, in the form of planar fractures and planar deformation features (PDFs) in quartz, is abundant throughout the Vredefort rocks, and provides evidence for shock pressures exceeding 30 GPa (refs 5, 6). Pseudotachylite breccias and impact melt dykes are ubiquitous throughout the crater. The shocked rocks contain original Archaean multidomain magnetite grains, as well as single-domain-sized magnetite particles crystallized within the PDFs[7]. High magnetic intensities in shocked granite-gneisses were attributed to the single-domain magnetite[8,9]. Paradoxically, the Earth's magnetic field is reduced by up to 4,000 nT at 300 m altitude over the area containing the highest magnetic intensities, which we will show is incompatible with shock demagnetization as suggested for the martian craters[2,3]. To explain this contradiction and the origin of the high remanence in these rocks, we carried out a palaeomagnetic study of the Vredefort crater.

We drilled three or more palaeomagnetic cores at 127 sites throughout the crater in the granite-gneisses, as well as in the pseudotachylites and impact dykes that cut them (Fig. 1). Each core was oriented with Sun and magnetic compasses. Strongly magnetized crust made Sun compass corrections necessary, as 22% of the measurements deviate more than 10° from the international geomagnetic reference field. Thermomagnetic experiments characterize Curie points of 577 ± 3 °C ($N = 3$) in the granite-gneisses and 571 ± 1 °C ($N = 4$) in the granulites, which correspond to Fe-pure magnetite and magnetite containing ~2% Ti, respectively.

One granite-gneiss sample, cut from one core per site, was subjected to stepwise alternating field (AF) demagnetization that normally isolated a very well defined component (Fig. 2a and b), whose direction was randomly oriented from one site to the next (Fig. 3a). Following these results, we thermally demagnetized samples from 31 cores, again isolating well defined linear magnetization components (Fig. 2c), with randomly distributed directions (Fig. 3b). The average angular distance between the magnetic vectors of two samples cut from two different cores from the same site is 64.7 ± 45.9°. As the average distance between two cores from the same site is 30 cm, the origin of the dispersion is less than a few decimetres.

To better define the length scale of the dispersion, we compared the magnetization directions between two samples cut from the same core for 34 cores (68 AF demagnetized samples). The average angular distance between magnetic vectors of the two samples from the same core is 9.4 ± 6.8° (omitting two outliers). This value remains high with respect to rocks possessing thermal remanent magnetizations (TRMs), which are rarely different by a degree, and suggests that the length scale of the dispersion lies within the dimension of a specimen (2.2 cm).

We then measured the anisotropy of magnetic susceptibility (AMS) on the same samples that underwent demagnetization. Interestingly, like the magnetic directions, principal susceptibility axes are randomly oriented on the site level (Fig. 3e), and the dispersion decreases as the length scale diminishes (Fig. 4). This is an important finding, as the magnetic foliations do not correlate with observable magmatic and metamorphic foliations, which trend consistently over distances greater than several metres. Because each sample contains multiple randomly oriented quartz grains that in turn contain multiple PDF geometries, the overall magnetic fabric is the sum of several individual fabrics that overprint the original magnetic foliation. If the AMS is controlled by single-domain magnetite, then the anisotropy of isothermal remanent magnetization (aIRM) should be defined by the same grains. Of nine aIRM experiments, the AMS and aIRM principal-axis directions are the same in seven samples whose anisotropies are high (9–29%) and similar in two cases with lower anisotropies (5%), meaning that the magnetite grains controlling the AMS also carry the magnetic remanence. Whereas most single-domain magnetite-bearing terrestrial rocks have inverse or intermediate AMS fabrics, Vredefort does not, owing to multiple individual planes of single-domain magnetite in the PDFs. Thus, the single-domain magnetite grains created during impact control the AMS and carry the magnetic remanence. Although lightning could potentially deviate the magnetization directions, it cannot reorient the magnetic anisotropy of the granite-gneisses. We conclude that the dispersion of the magnetization and AMS directions in the granite-gneisses is a characteristic feature of the impact event.

The magnetic behaviour is different for the melt rocks. A coherent direction is isolated at higher unblocking coercivities in 30 of 43 samples from five dykes (12 sites) distributed throughout the crater (declination $D = 25.0°$, inclination $I = 57.2°$, $\alpha_{95} = 3.9°$ (where α_{95} is the radius that the mean direction lies within 95% confidence); Figs 1, 2d and 3c). The remaining 13 samples, which come from the same dyke, have higher natural remanent magnetization (NRM) intensities (20.8 ± 5.8 A m^{-1}) than those possessing coherent directions (6.2 ± 5.7 A m^{-1}). The samples with high NRM intensities have possibly been hit by lightning. Nineteen of 23 pseudotachylite samples display a linear magnetization component (Fig. 2e), whose directions are well clustered within and

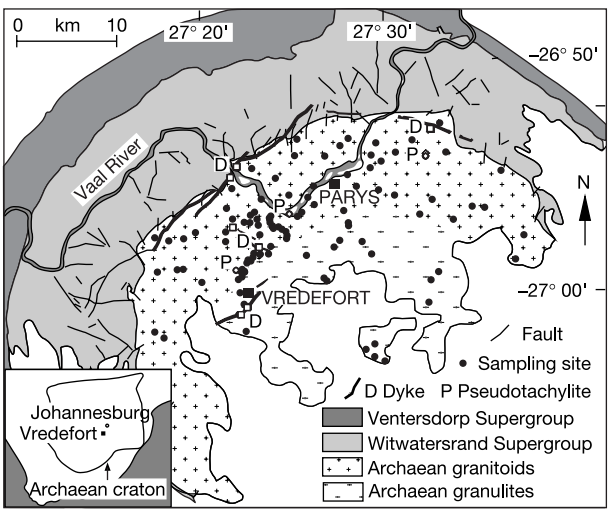


Figure 1 Simplified geologic map of the Vredefort crater (after ref. 25).

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between sites, defining a mean of $D = 20.8°$, $I = 57.4°$ ($\alpha_{95} = 1.9°$, Fig. 3d). The four pseudotachylite samples whose directions lie far from the dominant population have higher NRM intensities ($16.9 \pm 10.8\,A\,m^{-1}$) than the rest ($1.8 \pm 1.4\,A\,m^{-1}$). Again, we suspect that lightning is the cause for the aberrant directions in these latter samples.

Other palaeomagnetic studies in the Vredefort crater have also found consistent directions in the impact melt rocks[10,11], but basement rocks with high angular dispersions were rejected in the final analyses. Besides Vredefort, palaeomagnetic studies of other impact craters have found wildly variable magnetization directions in shocked facies[12,13]. Until now, no explanation has been given for the origin of these aberrant directions.

The shocked samples have extremely high NRM intensities and Koenigsberger (Q)-ratios (here defined as [NRM/(susceptibility × $23.9\,A\,m^{-1}$)]) compared with similar rock types found worldwide[14]. NRMs range from 0.01 to $110.0\,A\,m^{-1}$ (average, $16.2\,A\,m^{-1}$) and Q-ratios range from 0.2 to 748.7 (average, 107.7) for 182 samples. For comparison, NRM intensities and Q-ratios for the Cretaceous Mt Givens granite[15] (Sierra Nevada, California, $N = 247$) range from 0.01 to $7.3\,A\,m^{-1}$ (average, $0.4\,A\,m^{-1}$) and from 0.02 to 28.9 (average, 1.1); mean NRM and Q-ratios for the Archaean Pomovaara granite[16] (northern Finland) are $0.2\,A\,m^{-1}$ and 0.5 ($N = 141$). Because 26% of the dyke and pseudotachylite samples were probably struck by lightning, the same proportion of granites could have also been affected. Subtracting 26% of the samples with the highest intensities yields a range in NRMs from 0.01 to $22.0\,A\,m^{-1}$ (average, $5.7\,A\,m^{-1}$) and Q-ratios from 0.2 to 366.6 (average, 71.2) for $N = 135$; this subpopulation also has randomly oriented magnetic directions. We thus concur with Hart *et al.*[8] that the NRM intensities and Q-ratios of the Vredefort basement rocks are elevated compared to similar rock types found elsewhere.

Our study shows that impact acts to randomize and increase the magnetization in shocked rocks. The main difference between the shocked and non-shocked rocks is the relative timing of the lock-in of magnetic remanence. Pseudotachylites and dykes at Vredefort possess TRMs with magnetic characteristics similar to those of volcanic rocks that cooled in the Earth's magnetic field. Magnetic microscopy of the shocked facies identified preferential alignment of the magnetic moments within PDF planes[7], showing that there was an ambient magnetic field present when the magnetite grains acquired their remanences. However, because there is no hemispherical bias in the recorded palaeomagnetic directions (Fig. 3a and b), the external field direction could not have been constant over the crater. Thus, although PDFs control the distribution of magnetite in shocked rocks, an external field, whose orientation varied over centimetre length scales, must have been responsible for aligning their magnetic remanences. This is consistent with laboratory experiments, which show that plasma produced from impact events generates intense magnetic fields[17]. A 1-km-diameter bolide has been calculated to produce a 0.03-T field (1,000 times greater than

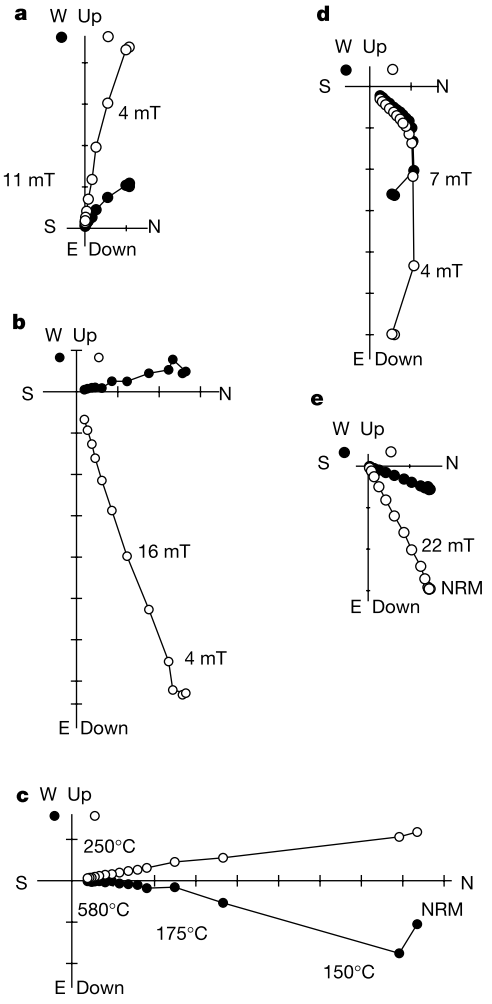


Figure 2 Representative demagnetization diagrams of Vredefort rocks. **a**, **b**, Alternating field (AF) demagnetization of shocked granite-gneiss; **a**, sample V4085A; **b**, V217A. **c**, Thermal demagnetization of shocked granite-gneiss (V047A). **d**, AF demagnetization of a dyke (V082B). **e**, AF demagnetization of a pseudotachylite (V4047A). Filled (open) symbols are the horizontal (vertical) magnetic component with 'up' and 'down' specifying inclination direction; N, S, E and W represent respectively north, south, east and west; NRM, natural remanent magnetization; specified demagnetization steps are in milliteslas (mT) or degrees Celsius (°C).

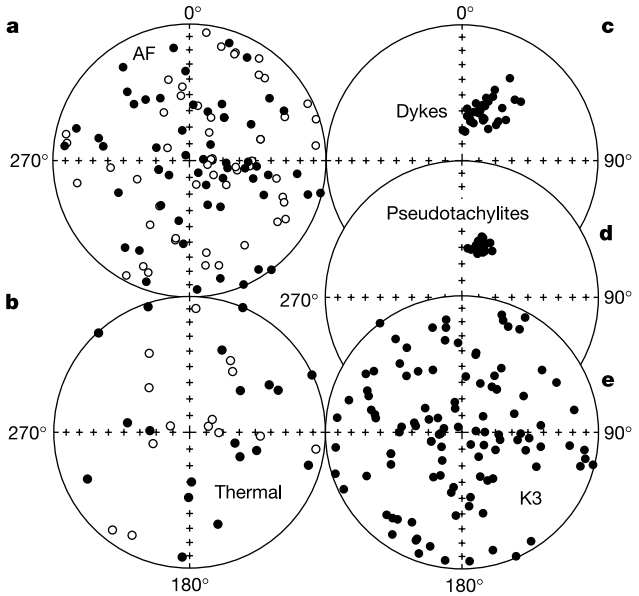


Figure 3 Equal area stereonet plots of the magnetic measurements. **a**, High coercivity magnetization component directions of 111 shocked granitoid samples. **b**, High temperature magnetization component of 31 shocked granitoid samples. **c**, High coercivity magnetization component of 30 dyke samples. **d**, High coercivity magnetization component of 23 pseudotachylite samples. **e**, Directions of minimum anisotropy of magnetic susceptibility axes (K3) of the same 111 shocked samples shown in **a**. Filled (open) circles are lower (upper) hemisphere projections of the magnetization component directions in **a** to **d**. In **e**, all minimum susceptibility axes directions are plotted on the lower hemisphere.

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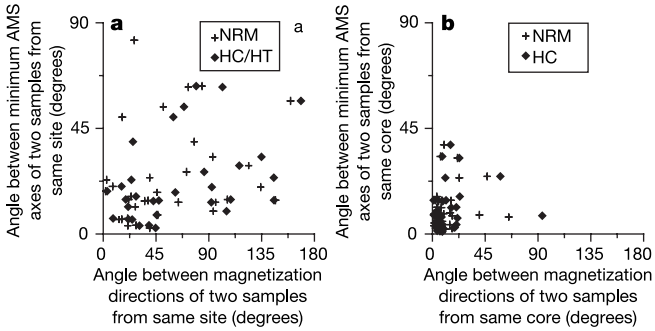


Figure 4 Comparison of the angular distance between minimum anisotropy of magnetic susceptibility (AMS) axes and magnetization component directions between two samples. **a**, Two samples from the same site; **b**, two samples from the same core. NRM, natural remanent magnetization direction; HC/HT, high coercivity or high temperature remanent magnetization direction; both are shown for comparison.

the Earth's) at a distance of 100 km that lasts for 100 s after an impact[18]. For Vredefort, these are minimum values because the meteorite diameter was greater than 10 km (ref. 19).

We suggest that small-wavelength magnetic fields of high intensity are responsible for both the randomization and the unusually high remanence of the Vredefort basement rocks. Diamond anvil experiments on magnetite show that its remanent saturation moment is significantly enhanced when it is either under pressure or after decompression[20]. This implies that shock creates a piezo-remanent magnetization, or a high-field TRM, depending on how fast the newly formed magnetite grains cooled through the Curie temperature, which is 580 °C at 0.1 MPa yet 1,180 °C at 30 GPa (refs 21, 22). Both processes would increase, not decrease, the magnetic intensities of the shocked rocks. Assuming that the temperature of the pre-impact crust was 500 °C, magnetite grains crystallizing in a 1-μm-thick PDF plane would cool through the Curie point in less than one second[7,23], in the plasma generated field. In contrast, a 1-m-wide dyke takes about 11 days to cool[23], long after the plasma field disappeared, in the presence of a 'normal' Earth field.

Aeromagnetic surveys over meteorite impacts commonly detect field intensities lower than regional magnetic trends[24]. Such is also the case for the gigantic (>1,000 km diameter) martian impact basins Hellas and Argyre[1], whose lower field intensities are commonly ascribed to pressure demagnetization[2,3]. Our study of Vredefort, the best terrestrial analogue of a martian meteorite crater, demonstrates that shock in fact leads to intensified and randomized magnetic vectors in shocked rocks. On the scale of the crater, the vectors cancel out, being equivalent to a volume of non-magnetic material seen at high altitudes, thus accounting for the reduced fields measured over the craters. The lack of crustal magnetism over impact basins on Mars is commonly ascribed to the absence of an internal dynamo during meteorite impact. Our new findings question the validity of this assumption. □

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1. Acuña, M. H. *et al.* Global distribution of crustal magnetization discovered by the Mars Global Surveyor MAG/ER experiment. *Science* **284,** 790–793 (1999).
2. Hood, L. L., Richmond, N. C., Pierazzo, E. & Rochette, P. Distribution of crustal magnetic fields on Mars: Shock effects of basin-forming impacts. *Geophys. Res. Lett.* **30,** 1281, doi:10.1029/2002GL016657 (2003).
3. Mohit, P. S. & Arkani-Hamed, J. Impact demagnetization of the Martian crust. *Icarus* **168,** 305–317 (2004).
4. Moser, D. E. Dating the shock wave and thermal imprint of the giant Vredefort impact, South Africa. *Geology* **25,** 7–10 (1997).
5. Stöffler, D. & Langenhorst, F. Shock metamorphism of quartz in nature and experiment: I. Basic observation and theory. *Meteoritics* **29,** 155–181 (1994).
6. Grieve, R. A. F., Langenhorst, F. & Stöffler, D. Shock metamorphism of quartz in nature and experiment: II. Significance in geoscience. *Meteoritics* **31,** 6–35 (1996).
7. Cloete, M. *et al.* Characterization of magnetite particles in shocked quartz by means of electron- and magnetic force microscopy: Vredefort, South Africa. *Contrib. Mineral. Petrol.* **137,** 232–245 (1999).
8. Hart, R. J., Hargraves, R. B., Andreoli, M. A. G., Tredoux, M. & Doucouré, C. M. Magnetic anomaly near the center of the Vredefort structure: implications for impact-related magnetic signatures. *Geology* **23,** 277–280 (1995).
9. Hart, R. J. *et al.* 'Super magnetic' rocks generated by shock metamorphism from the center of the Vredefort impact structure, South Africa. *S. Afr. J. Geol.* **103,** 151–155 (2000).
10. Hargraves, R. B. Paleomagnetic evidence relevant to the origin of the Vredefort ring. *J. Geol.* **78,** 253–263 (1970).
11. Jackson, G. M. *Paleomagnetic Study and Magnetic Modeling of the Vredefort Dome*. Thesis, Univ. Witwatersrand (1982).
12. Hargraves, R. B. & Roy, D. W. Paleomagnetism of anorthosite in and around the Charlevoix cryptoexplosion structure, Quebec. *Can. J. Earth Sci.* **11,** 854–859 (1974).
13. Halls, H. C. Shock-induced remanent magnetization in late Precambrian rocks from Lake Superior. *Nature* **255,** 692–695 (1975).
14. Clark, D. A. & Emerson, D. W. Notes on rock magnetic characteristics in applied geophysical studies. *Exp. Geophys.* **22,** 547–555 (1991).
15. Gilder, S. & McNulty, B. Tectonic exhumation and tilting of the Mount Givens pluton, central Sierra Nevada, California. *Geology* **27,** 919–922 (1999).
16. Wennerström, M. & Airo, M. L. Magnetic fabric and emplacement of the post-collisional Pomovaara Granite Complex in northern Fennoscandia. *Lithos* **45,** 131–145 (1998).
17. Crawford, D. A. & Schultz, P. H. Laboratory observations of impact-generated magnetic fields. *Nature* **336,** 50–52 (1988).
18. Crawford, D. A. & Schultz, P. H. Electromagnetic properties of impact-generated plasma, vapor and debris. *Int. J. Impact Eng.* **23,** 169–180 (1999).
19. Turtle, E. P. & Pierazzo, E. Constraints on the size of the Vredefort impact crater from numerical modeling. *Meteorit. Planet. Sci.* **33,** 483–490 (1998).
20. Gilder, S. A., LeGoff, M., Chervin, J. C. & Peyronneau, J. Magnetic properties of single and multi-domain magnetite under pressures from 0 to 6 GPa. *Geophys. Res. Lett.* **31,** L10612, doi:10.1029/2004GL019844 (2004).
21. Samara, G. & Giardini, A. Effect of pressure on the Néel temperature of magnetite. *Phys. Rev.* **186,** 577–580 (1969).
22. Schult, A. Effect of pressure on the Curie temperature of titanomagnetites [$(1-x)\cdot Fe_3O_4$–$x\cdot TiFe_2O_4$]. *Earth Planet. Sci. Lett.* **10,** 81–86 (1970).
23. Turcott, D. L. & Schubert, G. *Geodynamics* (Wiley & Sons, New York, 1982).
24. Grieve, R. A. F. & Pilkington, M. The signature of terrestrial impacts. *J. Austr. Geol. Geophys.* **16,** 399–420 (1996).
25. Nel, L. T. *Geological Map of the Country Around Vredefort. Scale 1:63360* (Union of S. Afr., Dept Mines and Ind., Geol. Surv., Pretoria, 1927).

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Advanced optics in a jellyfish eye

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Cubozoans, or box jellyfish, differ from all other cnidarians by an active fish-like behaviour and an elaborate sensory apparatus[1,2]. Each of the four sides of the animal carries a conspicuous sensory club (the rhopalium), which has evolved into a bizarre cluster of different eyes[3]. Two of the eyes on each rhopalium have long been known to resemble eyes of higher animals, but the function and performance of these eyes have remained unknown[4]. Here we show that box-jellyfish lenses contain a finely tuned refractive index gradient producing nearly aberration-free imaging. This demonstrates that even simple animals have been able to evolve the sophisticated visual optics previously known only from a few

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